TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Nanophotonic Chips for Rapid, Multiplexed Biomarker Testing*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		*Electrical・Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	*Molecular Genetic Diagnostic Devices*	*Optical Application Devices*	*Optical Materials*
Concept of Proposed Technology	*CMOS-compatible nanophotonic biosensor chip integrating high-Q silicon-metal resonator pixels for fast, label-free quantification and on-chip Raman hotspots for molecular fingerprint confirmation. Spatially barcoded capture probes enable multiplexed saliva or finger-stick blood panels; resonance shifts are read by camera imaging without a spectrometer, with on-chip references for calibration and drift correction.*			
Rationale for Joint R&D Project	*UIUC contributes device design, nano-fabrication, and multiplexed assay development. A Korean SME partner can drive manufacturability and commercialization through wafer-scale process integration, packaging and cartridges, reader optics/electronics, and pilot validation for the Korean market, enabling scaled production in Korea.*			
Keywords	*Nanophotonic biosensor chip; label-free multiplexing; silicon photonics; cavity-enhanced Raman; point-of-care diagnostics*			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (5)
R&D Objectives	*Existing capabilities:* *- Design and fabrication of dense silicon-metal nanophotonic pixel arrays with camera-readable intensity-shift transduction.* *- Spatial barcoding of capture probes (nucleic acids, proteins, and metabolites) plus on-chip controls and references for calibration and drift correction.* *R&D objectives for the joint project:* *- Co-integrated Raman hotspot pixels for label-free fingerprint confirmation of selected targets.* *- Improve manufacturability and yield using tolerance-aware design and wafer-scale process control.* *- Develop biofluid-stable, anti-fouling surface chemistry and multiplexed probe patterning with longitudinal stability.* *- Build an integrated cartridge + compact reader workflow suitable for non-expert operation.* *- Validate quantitative performance on representative saliva and finger-stick blood samples for a defined biomarker panel.*			
R&D Target Specifications	*Demonstrator target specifications (adjustable to the selected application with the SME partner):* *- Sample-to-answer time: 15 minutes or less.* *- Multiplexing: 20-50 targets per chip (expandable with array scaling).* *- Sensitivity: low-pM proteins and low-nM metabolites (application dependent).* *- Dynamic range: at least 10^4.* *- Repeatability: under 10% CV within-run and under 15% across days.* *- Specificity: false positive rate under 5% at decision thresholds using Raman confirmation on a subset of signature molecules.* *- Throughput: parallel imaging of hundreds to thousands of sensors per run; scalable to multi-million sensors per cm^2.* *- Sample volume: saliva 100-200 µL; finger-stick blood 20-50 µL (with minimal preprocessing).*			
R&D Plan	*Proposed joint R&D plan (24 months; adaptable to the TIPA program structure):* *Phase 1 (0-6 months): tolerance-aware pixel design and wafer process refinement; compact reader prototype (illumination + camera) with Raman channel separation; anti-fouling chemistry screening in saliva and plasma.* *Phase 2 (6-15 months): multiplex probe barcoding with internal controls; calibration and matrix-corrected quantification; disposable cartridge prototype and handling workflow validation.* *Phase 3 (15-24 months): integrated prototype (chip + cartridge + reader + software) and pilot validation; m anufacturing QC metrics, packaging and shelf-life tests; commercialization and regulatory pathway roadmap with the Korean SME.*			

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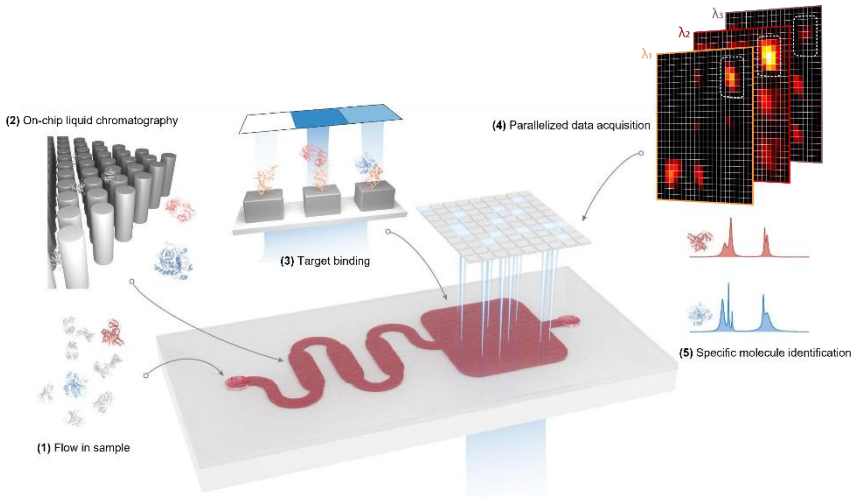
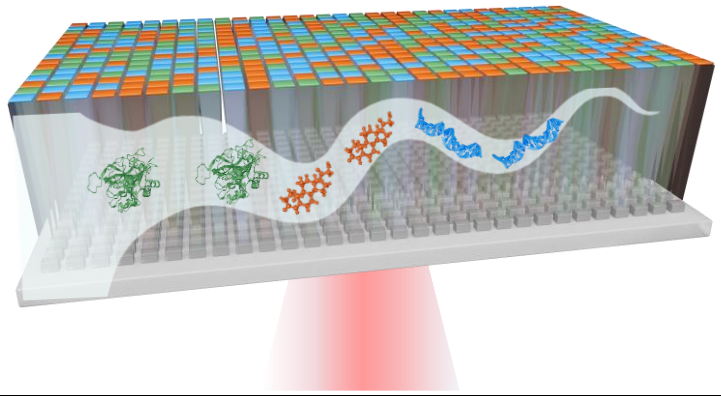

λ3
λ2
λ1
(2) On-chip liquid chromatography
(4) Parallelized data acquisition
(3) Target binding
(5) Specific molecule identification
(1) Flow in sample



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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Research Institute (UIUC) expected roles:* *- Lead nanophotonic chip design, fabrication, and optical characterization.* *- Develop and validate multiplexed assay chemistry, passivation, and spatial barcoding protocols.* *- Implement calibration and QC metrics, data processing, and quantitative interpretation logic.* *- Define verification test protocols and support system-level integration and troubleshooting.*
Expected R&R of Korean SME Partner	*Korean SME partner expected roles:* *- Co-develop manufacturable packaging and disposable cartridge design (materials, sealing, shelf life, user ergonomics).* *- Build or co-build reader hardware (optics, illumination, electronics) and user-facing software for field use.* *- Establish manufacturing process flow, QC, and supply chain for scaled production in Korea.* *- Support pilot testing and validation (including access to local partners for testing and market requirements), and lead commercialization planning for the Korean and global markets.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Expected outcomes:* *- A manufacturable nanophotonic biosensor chip and compact reader prototype with a validated multiplexed biomarker panel.* *- Disposable cartridge design with defined handling workflow and preliminary shelf-life data.* *- Wafer-level QC metrics, calibration strategy, and a verification package for repeatability and robustness.* *Expected impact and commercialization potential:* *- Enables rapid, quantitative, label-free point-of-care testing with higher information content than single-marker assays.* *- Creates a scalable reader-plus-disposable product pathway well-matched to Korean strengths in semiconductors, optics, and medical devices.* *- Provides a platform technology that can be reconfigured for additional applications (sports and workplace health monitoring, neurological care pathways, and other high-value diagnostic panels).*

6. References

1. Li, Y., Zhou, W., Tanriover. I., Hadibrata, W., Partridge, B.E., Lin, H., Hu, X., Lee, B., Dravid, V. P., Aydin, K., and Mirkin, C. A., Open Channel Metal Particle Superlattices, ***Nature***, 2022, 611, 695.

2. Zhou, W.*, Li, Y.*, Je, K., Vo, T., Lin, H., Partridge, B.E., Huang, Z., Glotzer, S., and Mirkin, C. A., Space-Tiled Colloidal Crystals from DNA-Forced Shape-Complementary Polyhedra Pairing, ***Science***, 2024, 383, 312.

3. Li, Y., Omo-Lamai, D., Dolia, V., Abdollahramezani, S., Moradifar, P., and Dionne, J. A. Multiplexed and Continu ous Monitoring of Metabolites using High-Quality Factor Metasurfaces and Modular DNA Aptamer Probes, 2026, submitted.

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5. Zhang, Y., Xu, D., Tanriover, I., Zhou, W., Li, Y., López-Arteaga, R., Aydin, K., Mirkin, C.A., Nonlinear Optical C olloidal Metacrystals. ***Nature Photonics***, 2024, 1-8.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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